

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm Examination

University Theory Examination November- December 2015

PH304 Pharmaceutical Biochemistry- I

Date: 8.12.2015, Tuesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION-I

Q 1 Attempt any TWO of the following

a. Define biochemistry. How it is useful in ordinary life, explain with suitable examples. **03**

A b. What are prokaryotes? How they differ from eukaryotes? Define various classes of prokaryotes based on their metabolic processes and give suitable examples. **07**

a. Explain various factors affecting the rate of enzyme activity. **07**

B b. What are enzyme inhibitors? Explain reversible enzyme inhibition with suitable examples. **03**

C Add a detailed note on "Regulation of Enzyme Inhibition in Living System" **10**

Q 2 Attempt any TWO of the following

a. Write down physiological role of following biomolecules:

A I) Lecithins II) Phosphatidyl-inositol III) Triacylglycerol **05**
IV) Steroids V) Eicosanoids

b. Explain the fatty acid Synthase enzyme complex. **05**

B Explain in detail cholesterol biogenesis and its regulation. **10**

a. Add a shortnote on ketogenesis and explain how they are useful for energy generation. **07**

C b. Draw structure of following fatty acids: **03**
I) 18:1;9 II) 20:4;5,8,11,14 III) 16:2;7,10

SECTION- II

Q 3 Attempt any TWO of the following

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| A | a. Describe HMP shunt and give its significance. | 05 |
| | b. Explain role of hormones in carbohydrates metabolism. | 05 |
| B | a. Describe metabolism of Phenylalanine and Tyrosine. | 05 |
| | b. Explain various secondary structures of protein. | 05 |
| C | a. Define carbohydrates. Classify them with examples. Give the name of various identification tests for carbohydrates. | 05 |
| | b. Write short notes on, I) Transamination II) Mitochondria | 05 |

Q 4 Attempt any TWO of the following

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| A | a. Explain TCA cycle along with energetics. | 06 |
| | b. Define: Glycogenesis, Gluconeogenesis, Deamination, Edman's reagent | 04 |
| B | a. Discuss the process of active transport across cell membrane. | 05 |
| | b. Add a short note on glycolysis. | 05 |
| C | a. Explain Krebs Henseleit's Cycle. | 05 |
| | b. Write short notes on, I) Protein structure identification II) Cori cycle | 05 |
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